Chammu Solar Plant

Generation Forecasting — how it works, and how to demo it

This document has two halves. The first explains what the system actually does with the plant's data — where every number on the screen comes from. The second is a step-by-step walkthrough for presenting it, including the honest talking points, because the fastest way to lose a room is to let it discover a limitation you did not mention.

Plant	Chammu — 12 inverters, 170 string combiner boxes, 6 meters
Data source	The plant's own SQL Server SCADA database. Nothing is typed in or estimated outside the system.
Record available	15 April – 19 July 2025. 42 complete days.
What is deployed	Generation forecasting only. The reporting modules are switched off for this site.
Status	Working and measured, but provisional — see section 3.

Part one — how it works

Read this before the demo so you can answer the questions without reaching for help.

1. Where the numbers come from

Every device in the plant writes to its own table in the SQL Server database — one row roughly every minute, carrying power, voltage, temperature and so on. The system reads those tables directly. It does not keep a second copy of the data and it never writes to them.

Turning readings into daily energy

A forecast needs energy per day, but the plant records power at an instant. Three things happen, in order:

- **De-duplicate.** The same minute sometimes appears more than once. Readings are rounded to the minute and averaged, so a minute logged three times counts once, not three times.

- **Integrate.** Each minute's power is worth one minute of energy, so the day's energy is the sum of the power readings divided by 60.

- **Reject incomplete days.** A day needs at least 1,380 of its 1,440 minutes. A day where the logger dropped out for two hours would look like a poor generation day and teach the model something false, so it is excluded rather than patched.

That last rule is why the system reports 42 usable days out of a longer calendar. The excluded days are not hidden — the Data Coverage panel shows the count.

A point specific to Chammu

Most plants have a PPC table recording total plant output in one place. Chammu does not. Plant generation here is built up from the 12 inverter tables instead. This is a real difference from other sites and it is worth knowing, because it means the plant figure is **gross generation at the inverters** — before transformer and cable losses, before auxiliary consumption. It is not the same thing as energy exported to the grid.

2. How a forecast is produced

The inputs

For each day the system builds a set of descriptive numbers: how much the plant generated on the previous days, the recent rolling average, the trend, the day of the year and where it sits in the seasonal cycle. Weather measurements from the on-site station are available as inputs — irradiance, module temperature, ambient temperature, wind, humidity, soiling.

> **Measured weather is deliberately not fed to the model as a predictor of the same day.** Using today's actual sunshine to predict today's output scores beautifully in testing and collapses in service, because in real operation you have a weather *forecast*, not a measurement. That shortcut is the most common way solar forecast accuracy gets inflated, and it was declined here. Connecting a licensed weather forecast feed is the biggest single improvement still available.

Choosing a model

Four candidate models compete on the same data, and the one that performs best on days it has never seen wins:

Model	What it does
Persistence	Predicts that tomorrow equals today. A genuine contender on a stable plant.
Seasonal Naive	Predicts from the same point in a recent repeating pattern.
Random Forest	Many decision trees, averaged. Handles non-linear behaviour.
Gradient Boosting	Trees built in sequence, each correcting the last.

The result is honest about itself: across the 10 inverters with enough history, Persistence won 3 times, Random Forest 3, Seasonal Naive 2 and Gradient Boosting 2. On a third of the devices the machine learning earns nothing over "the same as yesterday", and the system says so on screen rather than hiding it behind a sophisticated-sounding name.

How it is tested

Time-series data cannot be split at random. Shuffling the days would let the model learn from July to predict May, which is impossible in reality and produces a flattering, meaningless score. Instead the system walks forward in time: train on everything up to a date, predict the next day, step forward, repeat. The final block of days is held out entirely, scored once, and never used for tuning.

The uncertainty band

The shaded band around the forecast line is not a guess or a fixed percentage. It is built from the errors the model actually made on data it had never seen — the technique is called split-conformal prediction. If the model has been unreliable, the band is wide, automatically.

Why days 2 to 30 are different

Day one is predicted from real measured history. Day two is predicted using day one's *prediction* in place of a measurement, day three uses day two's, and so on. No new information enters the chain, and errors compound. This is why the 7-day and 30-day figures are labelled **PLANNING ESTIMATE** rather than treated as forecasts: they are a reasonable expected volume, not a tested prediction.

3. How accuracy is measured

Metric	Meaning	Current value
Score	100 minus WAPE. WAPE is total absolute error divided by total actual energy.	93.7% (plant, next day)

Metric	Meaning	Current value
Test days	How many unseen days the score is based on.	7
R2	Whether the model tracks day-to-day variation better than a flat average would.	**-1.64**
MAE / RMSE	Average error size, in MWh.	Shown on the accuracy row
Bias	Whether the model leans high or low overall.	Shown on the accuracy row

Read those two numbers together. 93.7% looks excellent and comfortably clears the 83% target. But it rests on seven held-out days, on a settled week where output barely moved — so predicting roughly the average was nearly right, which is largely what happened. The negative R2 confirms it: on those days the model tracked the ups and downs *worse* than a flat line would have. The system therefore refuses to mark the target as met, and shows an amber INSUFFICIENT DATA banner. Around 30 test days are needed before the score means anything. This is a data limitation, not a software defect.

4. Device-wise forecasting

The plant forecast answers "how much will the site make". The device section answers "which equipment is pulling its weight". They are deliberately separate calculations.

Category	Count	Forecast?	Why
Inverters	12	Yes — 10 of them	Each has its own power register, so energy can be integrated per device. Inverters 1 and 5 lack enough history and are left out rather than guessed.
Combiner boxes	170	No — ranked instead	No energy register, only DC power, and 34 days of it. They are compared against the median of their own group, which answers the more useful question anyway: which boxes are underperforming.
Meters	6	**No — blocked**	The export meter reads about 9% more energy than the inverters produce. Export cannot exceed generation, so one of the two is mis-scaled. Until that is resolved, meter forecasting stays off.

The per-inverter rows deliberately do **not** add up to the plant forecast, and the system says so on screen. Summing 10 of 12 inverters would understate the plant; scaling that sum up to cover the missing two would be inventing output that was never measured. Both were declined.

5. What the system will not do

- It will not claim the 83% target is met on seven test days.
- It will not use measured same-day weather to flatter the score.
- It will not scale 10 inverters up to 12.
- It will not quietly drop difficult days to improve the average.
- It will not forecast from meters whose readings do not reconcile.
- It will not state the cause of a deviation as fact — it attaches the evidence and labels the cause a hypothesis, for an engineer to confirm.

Each of those is a place where the number on screen could have been made to look better. They are worth mentioning in the demo, because they are the reason the numbers that *are* shown can be trusted.

Part two — the demo, step by step

Before they arrive (about 10 minutes)

Start the backend. Leave the terminal open.

```
cd C:\Users\servicescada32\Desktop\Chammu\Backend
.\.venv\Scripts\python.exe -m uvicorn app.main:app --host 127.0.0.1 --port 8001
```

Wait for `Uvicorn running on http://127.0.0.1:8001`. If you see `Errno 10048`, an old server is still holding the port — close it first.

Start the frontend in a second terminal.

```
cd C:\Users\servicescada32\Desktop\Chammu\Layout
npm.cmd run dev
```

- **Pre-run both forecasts.** Click *Run forecast* for the plant (a few seconds), then scroll down and click *Run device forecasts* (two to three minutes). Do this before the demo — nobody wants to watch a progress counter.

- **Open the Combiners tab once** so it caches. The first load scans 170 tables and takes about 20 seconds.

1 Open on the plant view

> *"This is Chammu — 12 inverters. Everything on this screen comes from the plant's own SCADA database. Nothing is typed in and nothing is estimated outside the system."*

2 The three volume cards

Point at **Next day**, **Next 7 days**, **Next 30 days** — tomorrow's expected generation, then the week, then the month. Then point at the badges, *without waiting to be asked*:

> *"Only the next-day figure is tested. The longer two say PLANNING ESTIMATE because beyond day one the model is working from its own previous prediction, with no new information arriving. Good for planning volumes, not a forecast."*

> Saying this first is the most important move in the demo. If the room discovers it ten minutes later, everything else you said comes into question.

3 The amber banner

> *"The model scored 93.7%, against a target of 83%. We are not calling that a pass, and here is why — seven held-out test days, on a settled week where output barely moved. We need around thirty before that number means anything."*

If they want the technical version: the R2 was negative, meaning the model tracked day-to-day variation worse than a flat average would have.

4 The chart

Green is actual, orange dashed is predicted, the blue band is the uncertainty range.

> *"Forty-two days of real generation. The band is built from the errors the model actually made on data it had never seen — so if it has been unreliable, the band widens by itself."*

5 Weather inputs

> *"No weather forecast feed is connected yet, so the model is predicting tomorrow purely from the shape of recent days. It has no idea whether tomorrow is cloudy. And deliberately, we do not feed it the weather we measured — using today's actual sunshine to predict today's output scores brilliantly in testing and falls apart in service, because in reality you have a forecast, not a measurement."*

This is the strongest credibility moment in the demo. It shows you turned down an easy number.

6 Deviations

> *"Once a predicted day finishes, we compare it with what happened and attach the evidence — irradiance, how many inverters were running. Causes are labelled as hypotheses, not conclusions. The measurement is a fact; the reason is for an engineer to confirm."*

7 Data coverage

> *"The record runs 15 April to 19 July 2025 — 42 complete days. Collection then stopped. That is the single biggest limitation here, and it is a data question rather than a software one."*

8 Scroll to the device section — Inverters tab

Point at the coverage panel first:

> *"Ten of twelve. Inverters 1 and 5 do not have enough history yet, and we do not scale up to cover them — so these rows deliberately do not add up to the plant number."*

Then the model column:

> *"Each inverter picks its own model. Three of them are best predicted by 'the same as yesterday'. The machine learning earns nothing there, and the system says so rather than hiding it."*

9 The finding that sells the work

Point at inverters **2, 6 and 10**: around 18 MWh against roughly 25 for the rest.

> *"Three inverters running about 30% below their peers, with identical rated power. That is roughly 22 MWh a day."*

> Then give the caveat, before anyone climbs on a roof: their internal module readings disagree with their main register in a way the healthy units do not. It could be a measurement fault rather than a panel fault. **Check the register first.**

10 Combiners tab

170 string combiner boxes, each compared against the median of its own group. Point at ICR1_SMB1 through SMB12: **0.0 kWh per day across 34 complete days.**

> *"Thirteen boxes in one section reading essentially nothing. They could be disconnected, uninstrumented, or working strings behind a dead sensor. We cannot tell which from here — but you now know exactly where to look."*

This is the most actionable thing on the screen, and it would be invisible at plant level.

11 Meters tab

> *"Six meters, forecasting deliberately switched off. The export meter reads about 9% more than the inverters produce, measured day by day across 39 days where both have data. Export cannot exceed generation, so one of them is mis-scaled — most likely a CT ratio. We have not adjusted either reading to make them agree, and we will not until someone confirms which one is right."*

Questions you will be asked

Question	Answer
Why isn't it 95% accurate?	It reads 93.7%, but on seven days. The number is not the problem — the evidence behind it is. Six months of data and we will know what it really is.
When can we rely on it?	Three to six months of continuous data for next-day forecasting. Connecting a weather forecast feed is the bigger lever, because cloud cover is what actually drives output.
Can you forecast each combiner box?	They have no energy measurement, only DC power, and 34 days of it. We rank them against each other instead, which answers the more useful question: which boxes are underperforming.
Can we trust the 30-day number?	As an expected volume for planning, yes. As a forecast, no — nothing has tested whether the model can predict 30 days out, and the card says so.
Why is the meter number higher than the inverters?	That is the open item. One of the two is mis-scaled. We have flagged it rather than papered over it, and meter forecasting stays off until it is settled.

Two things not to say

- Do not say "93.7% accurate". Say "scored 93.7% on a seven-day test, which we are treating as provisional".

- Do not say the three inverters are faulty. Say "reading low — worth investigating, and check the register before the panels".

What would move this forward

Restart data collection	Collection stopped on 19 July 2025. Nothing improves without more days.
Connect a weather forecast feed	The largest single accuracy gain available. Needs a licensed provider and the plant's coordinates.
Resolve the meter reconciliation	Confirm the CT/PT ratios. Unblocks all meter-side forecasting.
Check inverters 2, 6 and 10	Register first, then the panels. Roughly 22 MWh a day at stake.
Inspect ICR1_SMB1–12 and SMB16	Thirteen combiner boxes reporting zero.

Every figure in this document was measured from the Chammu database. Where something could not be measured, it is named as unmeasured rather than estimated.